

Aggregate Expenditure and Output in the Short Run

ECON 101 - Macroeconomics

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Fall 2026

- 8.1 The Aggregate Expenditure Model
- 8.2 Determining Aggregate Expenditure in the Economy
- 8.3 Graphing Macroeconomic Equilibrium
- 8.4 The Multiplier Effect
- 8.5 The Aggregate Demand Curve
- **Appendix** The Algebra of Macroeconomic Equilibrium

8.1 Learning Objective

Goal

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

- We build a simple short-run model called the **aggregate expenditure (AE) model**.
- It focuses on the relationship between:
 - total spending (**aggregate expenditure**) and
 - output (**real GDP**),
- assuming the **price level is constant** in the short run.

Aggregate Expenditure: Definition

- **Aggregate expenditure (AE):** total planned spending in the economy.
- Components:

$$AE = C + I + G + NX$$

- C : consumption
 - I : planned investment
 - G : government purchases
 - NX : net exports (exports minus imports)
- In this model we study how changes in these components affect equilibrium real GDP.

Four Components of Aggregate Expenditure

- **Consumption (C):**
 - Spending by households on goods and services.
- **Planned investment (I):**
 - Planned spending by firms on capital goods (machinery, buildings, equipment),
 - plus planned spending by households on new homes.
- **Government purchases (G):**
 - Government spending on goods and services at all levels (federal, provincial, local),
 - excluding transfer payments.
- **Net exports (NX):**
 - Value of exports minus the value of imports.

Planned Investment vs. Actual Investment

- National accounts measure **actual investment**.
- In the AE model we focus on **planned investment**:
 - Investment firms intend to carry out.
- **Actual investment** includes:
 - planned investment plus
 - **unplanned inventory changes** (changes in unsold goods).
- In the AE model:
 - Planned investment does not include unplanned changes in inventories.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium** occurs when:

$$\text{Planned aggregate expenditure} = \text{Real GDP}.$$

- If planned spending is greater than output:
 - firms sell more than expected,
 - inventories fall,
 - firms respond by increasing production and employment.
- If planned spending is less than output:
 - firms sell less than expected,
 - inventories rise,
 - firms respond by reducing production and employment.

Table 8.1 – Planned AE, GDP, and Inventories

Table 8.1 The Relationship between Planned Aggregate Expenditure and GDP

If ...	then ...	and ...
planned aggregate expenditure equals GDP,	inventories don't change	the economy is in macroeconomic equilibrium.
planned aggregate expenditure is less than GDP,	inventories rise	GDP and employment decrease.
planned aggregate expenditure is greater than GDP,	inventories fall	GDP and employment increase.

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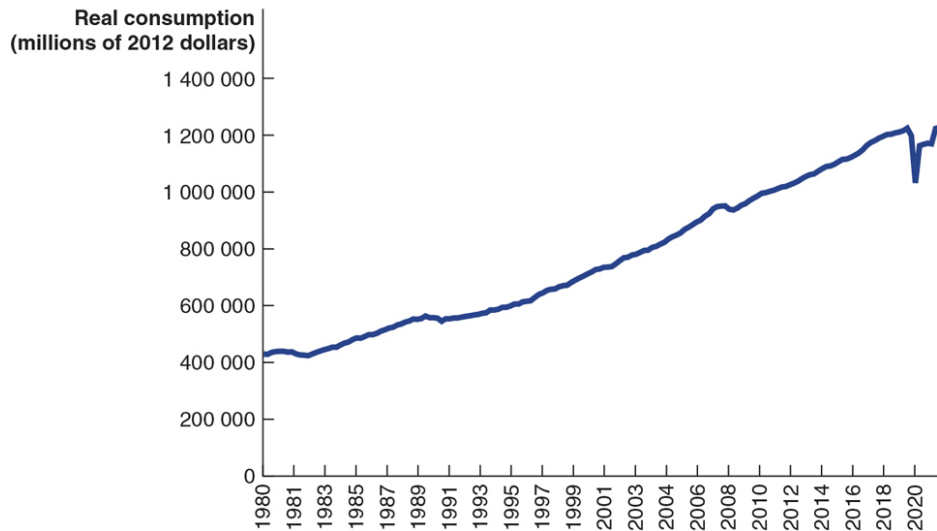
Table 8.2 Components of Aggregate Expenditure, 2021

SOURCE: Statistics Canada, “Gross Domestic Product, Expenditure-Based, Canada, Quarterly ($\times 1\,000\,000$),” Table 36-10-0104-01.

Expenditure Category	Real Expenditure (millions of 2012 dollars)
Consumption	1 227 356
Investment	378 117
Government Spending	530 077
Net Exports	−24 353

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Figure 8.1 – Real Consumption



- **Current disposable income**

- Main driver of consumer spending.
- Higher after-tax income $\Rightarrow$ higher consumption.

- **Household wealth**

- Wealth = assets (homes, stocks, bonds, bank accounts) minus liabilities (mortgages, loans).
- Wealthier households tend to spend more at any given income level.
- Empirical estimates: a permanent \$1 increase in wealth raises annual consumption by a few cents.

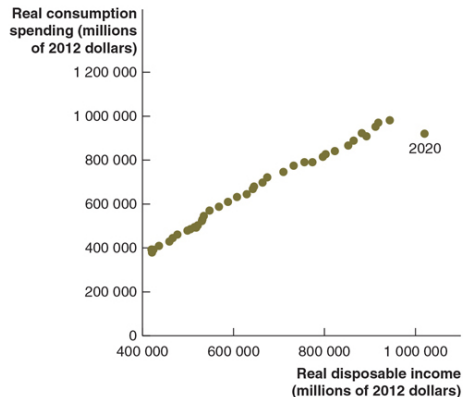
Determinants of Consumption (2 of 2)

- **Expected future income**
 - Households prefer **consumption smoothing**.
 - They base current spending partly on what they expect to earn in the future.
- **Price level**
 - Higher price level reduces the real value of wealth.
 - With the same dollar wealth, households can buy fewer goods and services $\Rightarrow$ lower consumption.
- **Interest rate**
 - Higher real interest rate encourages saving and discourages borrowing.
 - Especially reduces spending on durable goods (cars, appliances, houses).

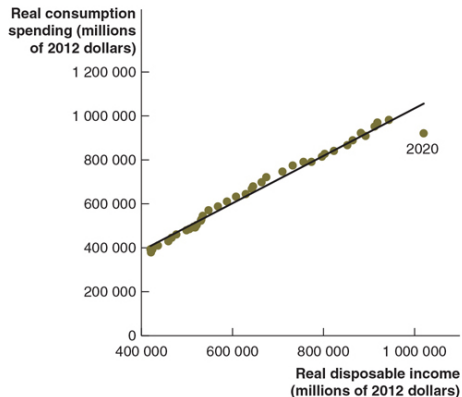
Consumption and Consumer Confidence

- Consumer spending also depends on how people feel about the future.
- **Consumer confidence** indices summarize:
 - expectations about income, employment, and overall economic conditions.
- When confidence rises:
 - households are more willing to spend,
 - AE shifts upward.
- When confidence falls:
 - households cut back on spending,
 - AE shifts downward.

Figure 8.2 – Consumption and Income, 1981–2020



(a) Relationship between consumption and income



(b) Solid line shows relationship between consumption and disposable income (consumption function)

The Consumption Function and MPC

- The **consumption function** describes how consumption depends on disposable income:

$$C = C_0 + \text{MPC} \times Y_D,$$

where:

- C_0 = autonomous consumption,
 - Y_D = disposable income,
 - MPC = marginal propensity to consume.
- **Marginal propensity to consume (MPC):**

$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}.$$

- It measures how much consumption changes when disposable income changes by one dollar.

MPC: Numerical Example

- Suppose data suggest that when disposable income rises by \$10,000:
 - consumption rises by about \$7,640.

- Then:

$$\text{MPC} \approx \frac{7,640}{10,000} = 0.764.$$

- Interpretation:
 - Households spend about **76.4%** of each additional dollar of income on consumption,
 - and save the remaining $\approx 23.6\%$.

- For this chapter we treat:

National income $\approx Y \approx$ real GDP.

- Disposable income:

$$Y_D = Y - T_{\text{net}},$$

where $T_{\text{net}} =$ taxes minus transfers.

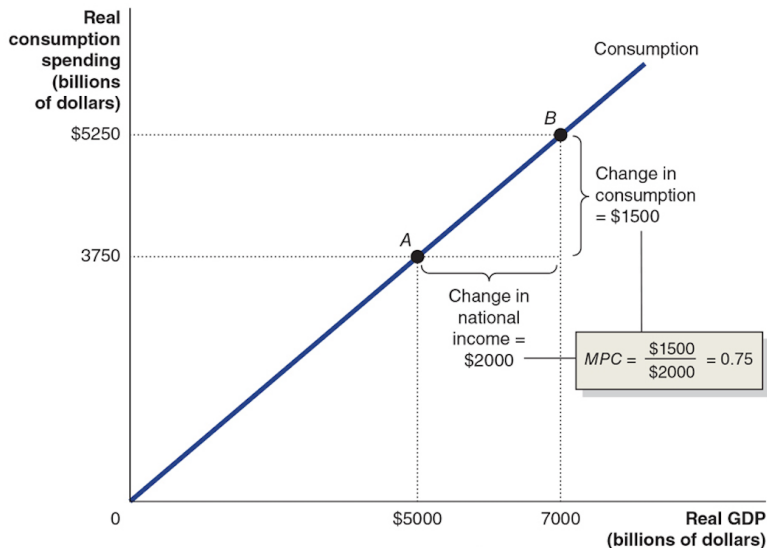
- Any part of disposable income not spent is **saving**:

$$Y_D = C + S.$$

Figure 8.3 – Consumption and National Income

National Income or GDP (billions of dollars)	Net Taxes (billions of dollars)	Disposable Income (billions of dollars)	Consumption (billions of dollars)	Change in National Income (billions of dollars)	Change in Disposable Income (billions of dollars)
\$ 1 000	\$1 000	\$ 0	\$ 750	—	—
3 000	1 000	2 000	2 250	\$2 000	\$2 000
5 000	1 000	4 000	3 750	2 000	2 000
7 000	1 000	6 000	5 250	2 000	2 000
9 000	1 000	8 000	6 750	2 000	2 000
11 000	1 000	10 000	8 250	2 000	2 000
13 000	1 000	12 000	9 750	2 000	2 000

Figure 8.3 – Consumption and National Income



Income Changes: Decomposing the Effects

- We can express changes in income as:

$$\Delta Y = \Delta C + \Delta S + \Delta T_{\text{net}}.$$

- We assume net taxes do not change with income in this simple model:

$$\Delta T_{\text{net}} = 0.$$

- Then:

$$\Delta Y = \Delta C + \Delta S.$$

- Dividing by ΔY :

$$1 = \frac{\Delta C}{\Delta Y} + \frac{\Delta S}{\Delta Y}.$$

- Marginal propensity to save (MPS):

$$\text{MPS} = \frac{\Delta S}{\Delta Y}.$$

- From the previous equation:

$$1 = \text{MPC} + \text{MPS}.$$

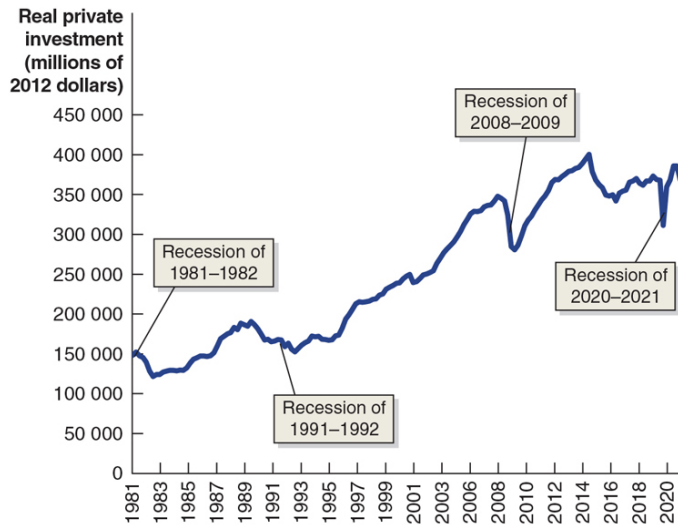
- So:

$$\text{MPS} = 1 - \text{MPC}.$$

- Interpretation:

- Every extra dollar of income is either:
 - spent (MPC part) or
 - saved (MPS part).

Figure 8.4 – Real Investment



Determinants of Planned Investment (1 of 2)

- **Expectations of future profits**
 - Capital goods and housing are long-lived.
 - Firms invest more when they are optimistic about future profitability.
 - Recessions and uncertainty reduce planned investment.
- Purchases of new housing (by households) are part of planned investment:
 - When household wealth falls or income expectations weaken, new housing investment falls.

Determinants of Planned Investment (2 of 2)

- **Real interest rate**

- Investment is often financed by borrowing.
- A higher real interest rate increases the cost of borrowing and reduces investment.
- A lower real interest rate encourages more investment.

- **Taxes**

- Higher corporate income taxes reduce after-tax profits and investment incentives.
- Investment tax credits can raise investment.

- **Cash flow**

- Many firms finance investment from internal funds (profits).
- Lower profits in recessions reduce cash flow and investment.

Figure 8.5 – Real Government Purchases

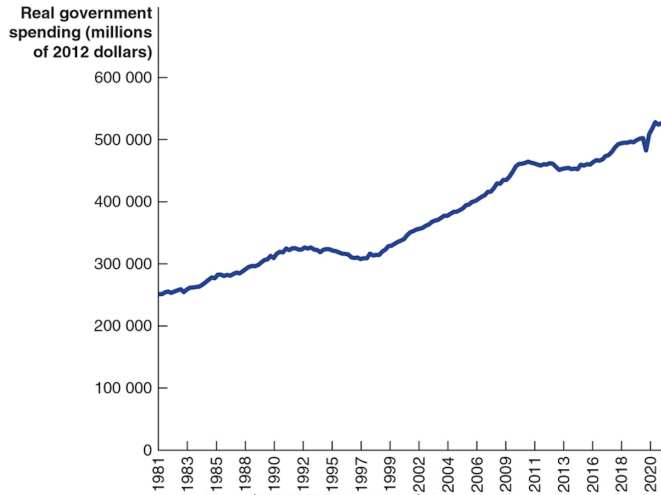
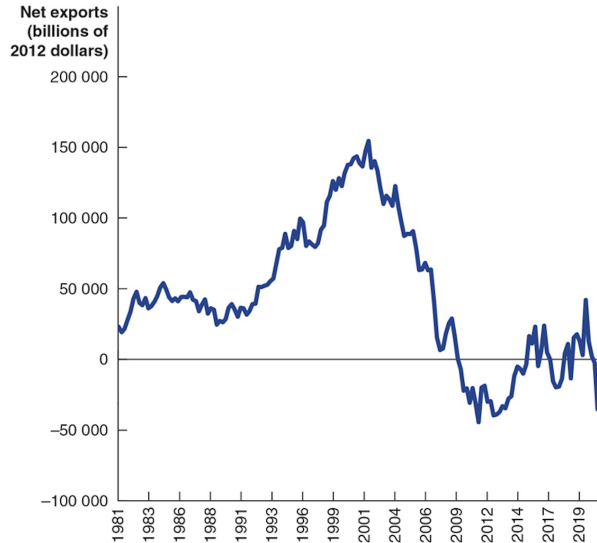


Figure 8.6 – Real Net Exports



Determinants of Net Exports

- Net exports depend on:
 - **Relative price levels**
 - If domestic prices rise faster than foreign prices, exports fall and imports rise.
 - **Relative growth rates**
 - If our economy grows faster than others, our imports tend to rise faster; exports may not keep up.
 - **Exchange rate**
 - A higher domestic currency value makes exports more expensive and imports cheaper $\Rightarrow$ lower net exports.

8.3 Learning Objective

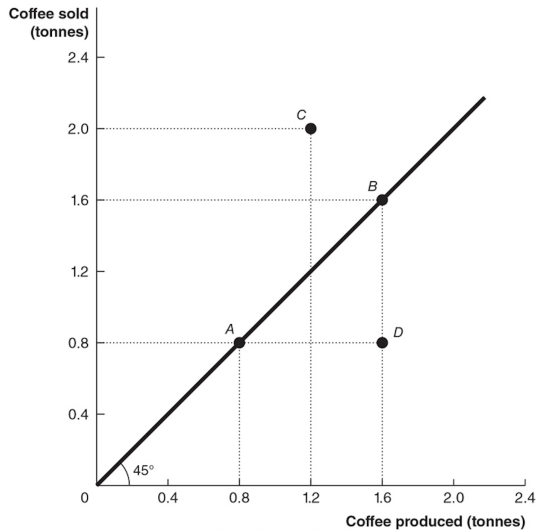
Goal

Use a 45°-line diagram to illustrate macroeconomic equilibrium.

- We graph real GDP on the horizontal axis and **planned aggregate expenditure** on the vertical axis.
- The 45° line shows all points where:

$$AE = Y.$$

Figure 8.7 – The 45°-Line Diagram



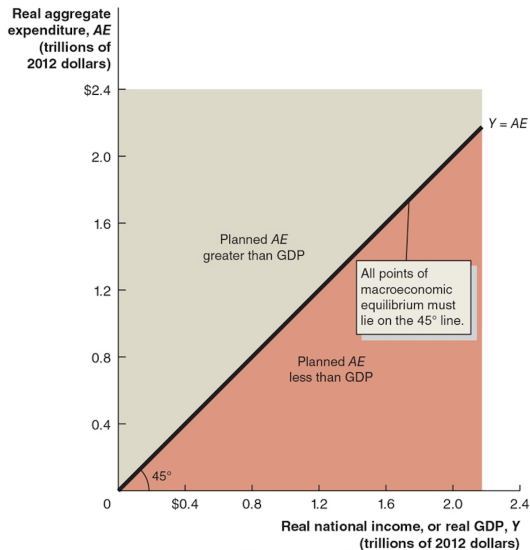
Understanding the 45° Line

- On the 45° line:

$$\text{Real GDP} = \text{Planned AE.}$$

- At a point like **C** (above 45° line):
 - Planned AE exceeds GDP,
 - inventories fall,
 - firms increase production $\Rightarrow$ GDP rises toward equilibrium.
- At a point like **D** (below 45° line):
 - Planned AE is less than GDP,
 - inventories rise,
 - firms cut production $\Rightarrow$ GDP falls toward equilibrium.

Figure 8.8 – AE and GDP in the 45° Diagram



- AE is built from its components:

$$AE = C(Y) + I + G + NX.$$

- In this short-run model:
 - C increases with Y ,
 - I, G, NX are taken as **autonomous** (do not change with income).
- Graphically, AE is an upward-sloping line because:
 - higher income $\Rightarrow$ higher consumption $\Rightarrow$ higher AE.

Figure 8.9 – Finding Macroeconomic Equilibrium

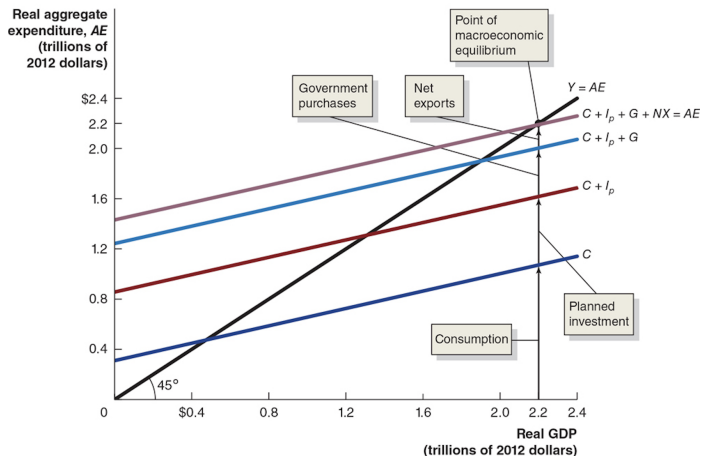


Figure 8.10 – Macroeconomic Equilibrium

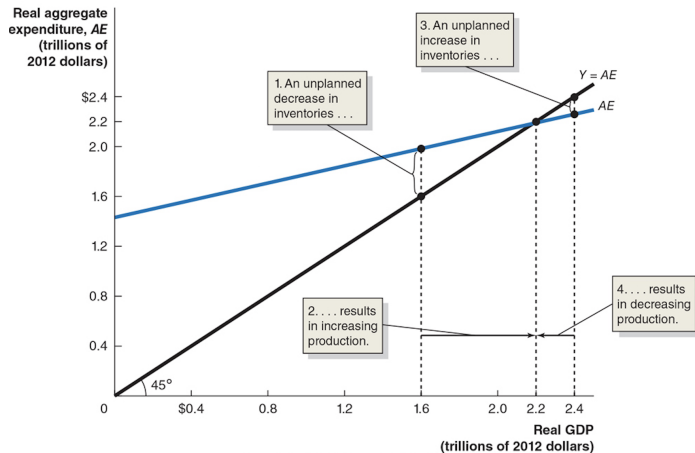
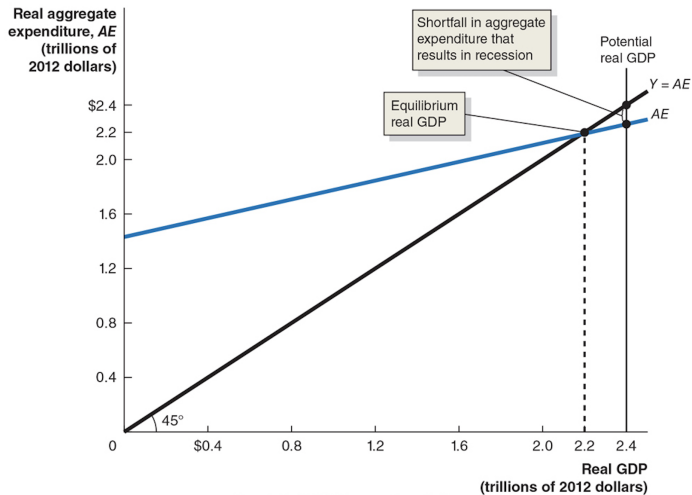


Figure 8.11 – A Recession in the 45° Diagram



Inventories and Adjustment

- Inventories provide the **signal** that triggers movement toward equilibrium.
- If planned $AE < Y$:
 - firms accumulate unplanned inventory,
 - they cut production and employment.
- Even when spending returns to normal, firms may first run down the stock of inventories before increasing output.
- If planned $AE > Y$:
 - inventories are depleted,
 - firms increase output and hire more workers.

Table 8.3 – Macroeconomic Equilibrium

Table 8.3 Macroeconomic Equilibrium

Real GDP (Y)	Consumption (C)	Planned Investment (I_p)	Government Purchases (G)	Net Exports (NX)	Planned Aggregate Expenditure (AE)	Unplanned Inventory Investment	Real GDP Will . . .
\$800	\$740	\$125	\$125	-\$30	\$960	-\$160	increase
1200	1060	125	125	-30	1280	-80	increase
1600	1380	125	125	-30	1600	0	be in equilibrium
2000	1700	125	125	-30	1920	80	decrease
2400	2020	125	125	-30	2240	160	decrease

NOTE: The values are in billions of 2012 dollars.

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8.4 Learning Objective

Goal

Describe the multiplier effect and use the multiplier formula to calculate changes in equilibrium GDP.

- A small change in **autonomous expenditure** causes a larger change in equilibrium real GDP.
- This amplification is called the **multiplier effect**.

Autonomous vs. Induced Expenditure

- **Autonomous expenditure:**
 - Parts of spending that do not depend on the level of real GDP.
 - Examples: I , G , NX , and the autonomous part of C .
- **Induced expenditure:**
 - Spending that *does* depend on real GDP.
 - For us, this is the part of consumption that changes with income.
- Because consumption rises with income, an increase in autonomous spending sets off a chain reaction of additional spending.

Figure 8.12 – The Multiplier Effect

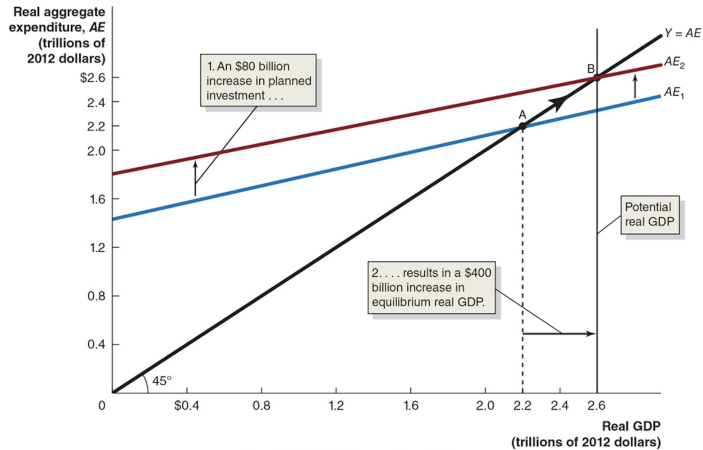


Table 8.4 – The Multiplier in Action

Table 8.4 The Multiplier Effect in Action

	Additional Autonomous Expenditure	Additional Induced Expenditure (C)	Total Additional Expenditure = Total Additional GDP
Round 1	\$80 billion	\$0	\$80.00 billion
Round 2	0	64.00 billion	144.00 billion
Round 3	0	51.20 billion	195.20 billion
Round 4	0	40.96 billion	236.16 billion
Round 5	0	32.77 billion	268.93 billion
.	.	.	.
.	.	.	.
.	.	.	.
Round 10	0	10.74 billion	357.05 billion
.	.	.	.
.	.	.	.
.	.	.	.
Round 20	.	1.15 billion	395.39 billion
.	.	.	.
Round n	0	0	400.00 billion

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Eventual Effect of the Multiplier

- Suppose an initial \$80 billion increase in planned investment.
- First round:
 - AE rises by \$80 billion, so firms increase production by \$80 billion.
- Second round:
 - Income is now higher by \$80 billion.
 - Households consume $MPC \times 80$ billion more.
- Third round:
 - The extra consumption becomes income again, leading to further consumption, and so on.
- Eventually, equilibrium GDP rises by **more** than \$80 billion.

Multiplier and MPC: Geometric Series

- Total change in equilibrium GDP:

$$\Delta Y = \Delta A + (\text{MPC})\Delta A + (\text{MPC})^2\Delta A + (\text{MPC})^3\Delta A + \dots$$

- Factor out ΔA :

$$\Delta Y = \Delta A [1 + \text{MPC} + (\text{MPC})^2 + \dots].$$

- The term in brackets is an infinite geometric series:

$$1 + \text{MPC} + (\text{MPC})^2 + \dots = \frac{1}{1 - \text{MPC}} \quad (\text{since } 0 < \text{MPC} < 1).$$

Formula for the Multiplier

- Therefore:

$$\Delta Y = \frac{1}{1 - \text{MPC}} \times \Delta A.$$

- The **multiplier** is:

$$\text{Multiplier} = \frac{1}{1 - \text{MPC}}.$$

- Example:

- If $\text{MPC} = 0.8$,
- multiplier $= \frac{1}{1-0.8} = 5$.
- An \$80 billion increase in investment raises equilibrium GDP by:

$$\Delta Y = 5 \times 80 = 400 \text{ billion.}$$

Summarizing the Multiplier Effect

- The multiplier works in both directions:
 - Increases in autonomous expenditure raise GDP by a multiple of the initial change.
 - Decreases in autonomous expenditure lower GDP by a multiple of the initial change.
- The larger the **MPC**, the larger the multiplier.
- Our simple model:
 - ignores some leakages (e.g., imports, taxes, interest-rate and price changes),
 - so real-world multipliers are usually smaller than this simple formula suggests.

The Paradox of Thrift

- Long run:
 - Higher saving can finance more investment and foster growth.
- Short run (in the AE model):
 - If households decide to save more (and consume less),
 - consumption falls, AE falls,
 - equilibrium GDP falls, possibly causing a recession.
- This tension is known as the **paradox of thrift**:
 - What is prudent for individuals can be harmful for the economy in the short run.
- In practice, changes in saving also affect interest rates and investment, so the size of this effect is an empirical question.

8.5 Learning Objective

Goal

Understand the relationship between the aggregate demand curve and aggregate expenditure.

- So far we treated the price level as constant.
- Now we ask: what happens to AE when the **price level changes**?

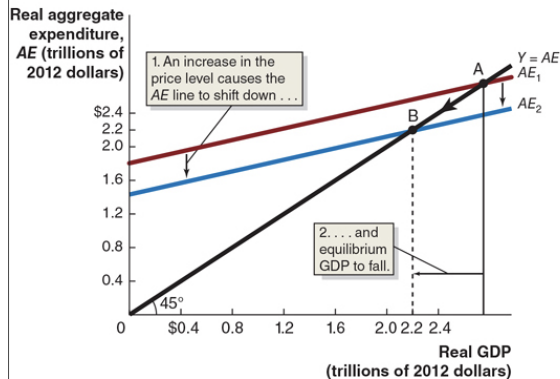
Price Level and Aggregate Expenditure

- A higher overall price level:
 - reduces the real value of household wealth,
 - makes domestic goods more expensive relative to foreign goods,
 - tends to raise interest rates when the money supply is fixed.
- These channels reduce:
 - consumption,
 - investment,
 - and net exports.
- Therefore, a higher price level $\Rightarrow$ lower aggregate expenditure.
- A lower price level has the opposite effects.

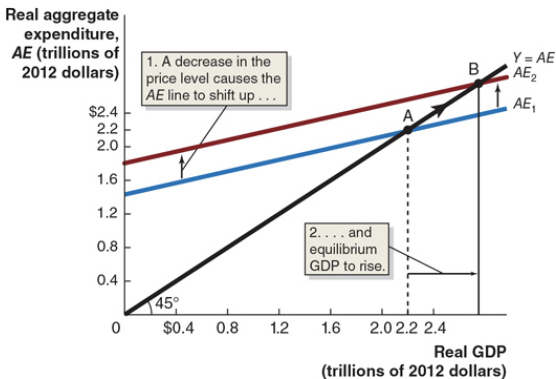
How the Price Level Affects AE

- **Wealth effect (on consumption):**
 - Higher price level reduces real wealth, decreasing consumption.
- **International trade effect (on net exports):**
 - If domestic prices rise faster than foreign prices, exports fall and imports rise, reducing net exports.
- **Interest rate effect (on investment):**
 - Higher price level increases money demand.
 - With a fixed money supply, interest rates rise, reducing investment.

Figure 8.13 – Price Level Changes and Real GDP



(a) The effect of a higher price level on real GDP



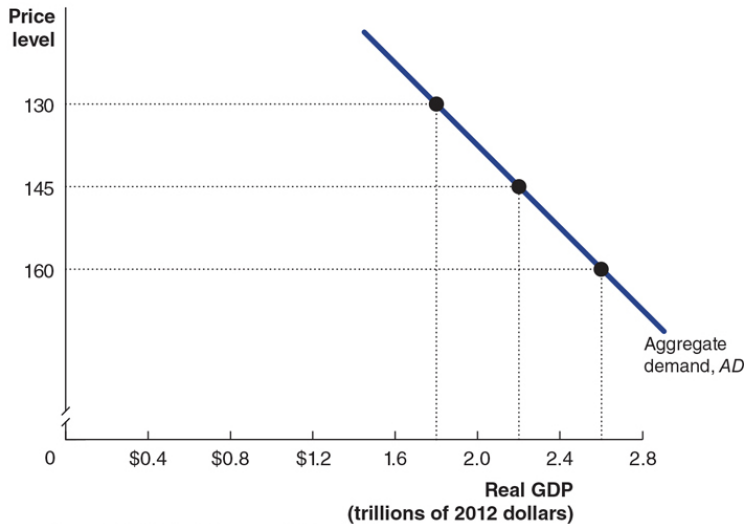
(b) The effect of a lower price level on real GDP

From AE to the Aggregate Demand Curve

- For each possible price level, we can:
 - draw an AE line,
 - find the corresponding equilibrium real GDP.
- Plotting the resulting pairs (price level, equilibrium GDP) gives the:
 - **aggregate demand (AD) curve.**
- **Aggregate demand curve:**
 - relationship between the price level and the quantity of real GDP demanded,
 - holding constant other factors that affect AE.

Figure 8.14 – The Aggregate Demand Curve

Price level	Equilibrium real GDP
130	\$2.6 trillion
145	2.2 trillion
160	1.8 trillion



Common Misconceptions

- In the 45°-line diagram:
 - **Real GDP (Y)** and **aggregate expenditure (AE)** are conceptually different.
 - They are equal only at macroeconomic equilibrium.
- **Consumption** is only one component of AE:
 - AE includes $C + I + G + NX$.
- The aggregate demand curve:
 - shows the effect of the price level on equilibrium GDP,
 - not just on consumption.

Goal

Apply the algebra of macroeconomic equilibrium to obtain numerical results.

- Graphs show the **direction** of change.
- Algebraic models allow us to compute **numerical** changes.

Aggregate Expenditure Equations: Example

- A simple numerical AE model might look like:

$$C = \bar{C} + \text{MPC} Y,$$

$$I = \bar{I},$$

$$G = \bar{G},$$

$$NX = \bar{NX},$$

$$Y = C + I + G + NX.$$

- The bars indicate **autonomous** values (do not change with income).
- MPC is the marginal propensity to consume from national income.

Solving the Model: Example

- Suppose:

$$C = 100 + 0.8Y,$$

$$I = 125,$$

$$G = 125,$$

$$NX = -30.$$

- Then the equilibrium condition:

$$Y = C + I + G + NX$$

becomes:

$$Y = 100 + 0.8Y + 125 + 125 - 30.$$

- Rearranging:

$$Y - 0.8Y = 100 + 125 + 125 - 30 \Rightarrow 0.2Y = 320.$$

- So:

$$Y = \frac{320}{0.2}$$

$$= 1600$$

General Form and Interpretation

- In general we can write:

$$\begin{aligned}C &= \bar{C} + cY, \\I &= \bar{I}, \quad G = \bar{G}, \quad NX = \bar{NX}, \\Y &= C + I + G + NX.\end{aligned}$$

- Substituting:

$$Y = \bar{C} + cY + \bar{I} + \bar{G} + \bar{NX}.$$

- Rearranging:

$$Y - cY = \bar{C} + \bar{I} + \bar{G} + \bar{NX} \quad \Rightarrow \quad (1 - c)Y = A,$$

where A is total autonomous expenditure.

- Therefore:

$$Y = \frac{1}{1 - c} \times A = \text{Multiplier} \times \text{Autonomous expenditure}.$$

- The **AE model** explains short-run equilibrium when the price level is fixed.
- Equilibrium occurs where:

$$\text{Planned AE} = Y.$$

- **Consumption** is driven by disposable income, wealth, expectations, prices, and interest rates.
- **Investment, government purchases, and net exports** are treated as autonomous in the basic model.
- The **multiplier effect** makes equilibrium GDP more sensitive to changes in autonomous expenditure.
- Changes in the **price level** affect AE through wealth, international trade, and interest rate channels, generating a downward-sloping **aggregate demand curve**.

Thank you!